

2 Continuous probability distributions	2.1	The concept of a continuous random variable. The probability density function and the cumulative distribution function for a continuous random variable.	Students will be expected to link with their knowledge of histograms and frequency polygons. Use of the probability density function $f(x)$, where $P(a < X \leq b) = \int_a^b f(x) \, dx$ Use of the cumulative distribution function $F(x_0) = P(X \leq x_0) = \int_{-\infty}^{x_0} f(x) \, dx .$ The formulae used in defining $f(x)$ and the calculus required will be consistent with the level expected in A level Mathematics and A level Further Mathematics.
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Topics	What students need to learn:		
	Content		Guidance
2 Continuous probability distributions <i>continued</i>	2.2	Relationship between probability density and cumulative distribution functions.	$f(x) = \frac{dF(x)}{dx}.$
	2.3	Mean and variance of continuous random variables. Extension of expected value function to include $E(g(X))$ Mode, median and percentiles of continuous random variables. Idea of skewness	The formulae used to define $g(x)$ will be consistent with the level required in A level Mathematics and A level Further Mathematics Questions may require candidates to use these calculations to assess the suitability of models. Candidates will be expected to describe the skewness as positive, negative or zero and give a suitable justification.
	2.4	The continuous uniform (rectangular) distribution.	Including the derivation of the mean, variance and cumulative distribution function.